

Pharmaceutical Reform and Pharmacy Entry: Evidence from Hospitals in China

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Data Sources

Hospital Data. Information on hospitals is obtained from business registration records, including hospital name, classification (level), industry code, province, county, and geographic coordinates.

Pharmacy Data. Pharmacy data are collected from Dianping, including merchant name, industry classification, geographic coordinates, and an indicator for chain affiliation.

Reform Data. Information on the timing and implementation of the reform is compiled from official government policy documents and related reports.

Nighttime Light Data. VIIRS nighttime light (NTL) data are extracted from Google Earth Engine, capturing average luminosity within 1km, 2km, and 3km buffers around each hospital.

Descriptive Statistics

Variable	Mean	SD	P25	P50	P75
<i>Panel A. Pharmacy outcomes</i>					
Pharmacy Count (1km)	10.898	11.800	2	7	17
Pharmacy Count (2km)	30.820	31.416	5	22	46
Pharmacy Count (3km)	53.411	55.518	10	37	79
Log Pharmacy (1km)	1.897	1.192	1.099	2.079	2.890
Log Pharmacy (2km)	2.770	1.405	1.792	3.135	3.850
Log Pharmacy (3km)	3.249	1.499	2.398	3.638	4.382
Has Pharmacy (1km)	0.820	0.384	1	1	1
Has Pharmacy (2km)	0.894	0.308	1	1	1
Has Pharmacy (3km)	0.920	0.271	1	1	1
<i>Panel B. Nighttime lights and treatment status</i>					
Log NTL Mean (1km)	2.679	1.000	2.082	2.896	3.454
Log NTL Mean (2km)	2.531	1.013	1.885	2.706	3.339
Log NTL Mean (3km)	2.394	1.021	1.682	2.520	3.223
DID	0.291	0.454	0	0	1
Treat	0.338	0.473	0	0	1
Post	0.806	0.395	1	1	1

Empirical Strategy: TWFE

$$Y_{ict} = \beta_0 + \beta_1 Reform_{ct} + \beta_2 NTL_{ict} + \delta_i + \delta_t + \epsilon_{ict}$$

- ▶ Y_{ict} : Number of pharmacies around hospital i in county c at time t
- ▶ $Reform_{ct}$: Policy exposure at the county-year level
- ▶ NTL_{ict} : Nighttime light intensity (proxy for local economic activity)
- ▶ δ_i : Hospital fixed effects
- ▶ δ_t : Year fixed effects

Empirical Strategy: CS-DID

$$ATT(g, t) = \mathbb{E}[Y_t(1) - Y_t(0) \mid G = g]$$

- ▶ g : First year of reform (cohort)
- ▶ t : Year
- ▶ Compare treated units with not-yet-treated units
- ▶ Allows for heterogeneous treatment effects
- ▶ Avoids bias from two-way fixed effects under staggered adoption

Estimate dynamic and cohort-specific treatment effects

Sensitivity to Fixed Effects and Controls (2km)

	(1)	(2)	(3)	(4)	(5)
DID	0.800*** (0.167)	-2.465** (0.773)	0.527 (0.441)	0.520 (0.441)	0.062 (0.398)
Year FE	No	Yes	Yes	Yes	Yes
Hospital FE	No	No	Yes	Yes	Yes
NTL Control	No	No	No	Yes	Yes
Province Trend	No	No	No	No	Yes
N	166,113	165,909	164,703	164,674	164,674
R ²	0.000	0.040	0.905	0.905	0.910

Sensitivity to Fixed Effects and Controls (Log Outcome, 2km)

	(1)	(2)	(3)	(4)	(5)
DID	0.181*** (0.007)	-0.043 (0.032)	0.004 (0.018)	0.007 (0.018)	0.003 (0.017)
Year FE	No	Yes	Yes	Yes	Yes
Hospital FE	No	No	Yes	Yes	Yes
NTL Control	No	No	No	Yes	Yes
Province Trend	No	No	No	No	Yes
N	166,113	165,909	164,703	164,674	164,674
R ²	0.003	0.111	0.919	0.919	0.923

Baseline DID Results: TWFE (Level)

- ▶ Outcome: Pharmacy count around hospitals
- ▶ Fixed effects: Hospital and year
- ▶ Standard errors: clustered at county level

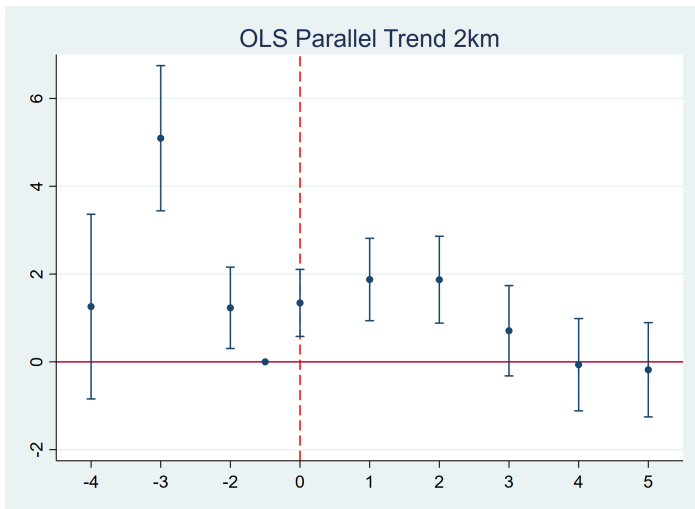
	<i>Baseline</i>			<i>+ Nighttime Lights</i>		
	(1) 1km	(2) 2km	(3) 3km	(4) 1km	(5) 2km	(6) 3km
DID	0.065 (0.169)	0.527 (0.441)	1.632* (0.784)	0.062 (0.169)	0.520 (0.441)	1.609* (0.782)
Hospital FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
NTL Control	No	No	No	Yes	Yes	Yes
N	164,703	164,703	164,703	164,674	164,674	164,674
R ²	0.895	0.905	0.910	0.895	0.905	0.910

Baseline DID Results: TWFE (Log)

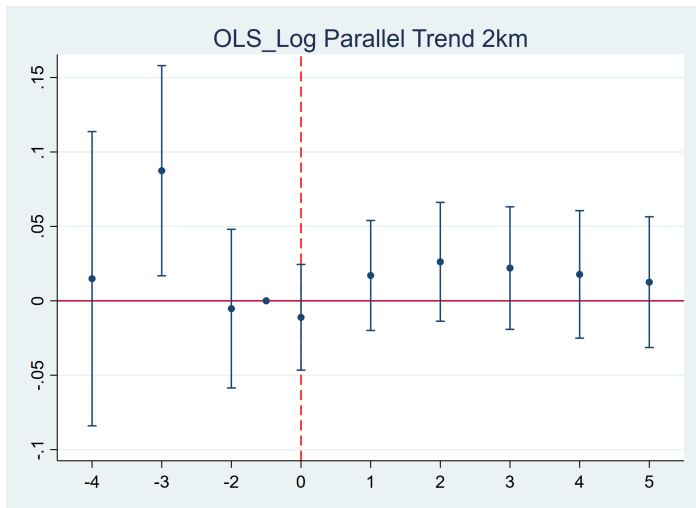
- ▶ Outcome: Log of pharmacy count around hospitals
- ▶ Fixed effects: Hospital and year
- ▶ Standard errors: clustered at county level

	<i>Baseline</i>			<i>+ Nighttime Lights</i>		
	(1) 1km	(2) 2km	(3) 3km	(4) 1km	(5) 2km	(6) 3km
DID	0.004 (0.016)	0.004 (0.018)	-0.004 (0.020)	0.005 (0.016)	0.007 (0.018)	0.000 (0.019)
NTL				0.051*** (0.015)	0.153*** (0.022)	0.244*** (0.029)
Hospital FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
NTL Control	No	No	No	Yes	Yes	Yes
N	164,703	164,703	164,703	164,674	164,674	164,674
R ²	0.908	0.919	0.922	0.908	0.919	0.923

Event Study: TWFE (Level)



Event Study: TWFE (Log)



CS-DID: Counts Level

Callaway and Sant'Anna (2021)

- ▶ Control: never treated
- ▶ Standard errors: clustered at county level

	(1) pharmacy~1km	(2) pharmacy~2km	(3) pharmacy~3km
ATT	0.394*** (0.118)	1.112*** (0.301)	2.298*** (0.524)
N	110,087	110,087	110,087

CS-DID: Log Level

Callaway and Sant'Anna (2021)

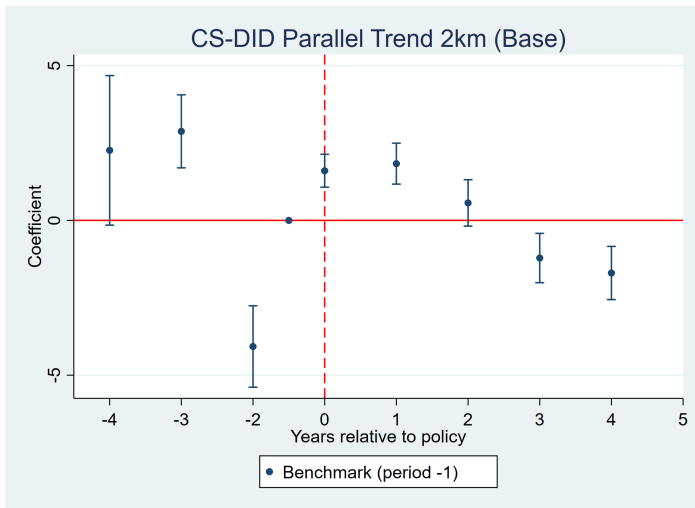
- ▶ Outcome: significant positive
- ▶ Control: never treated

	(1) ln(pharmacy~1km)	(2) ln(pharmacy~2km)	(3) ln(pharmacy~3km)
ATT	0.019 (0.012)	0.026* (0.013)	0.025* (0.014)
N	110,087	110,087	110,087

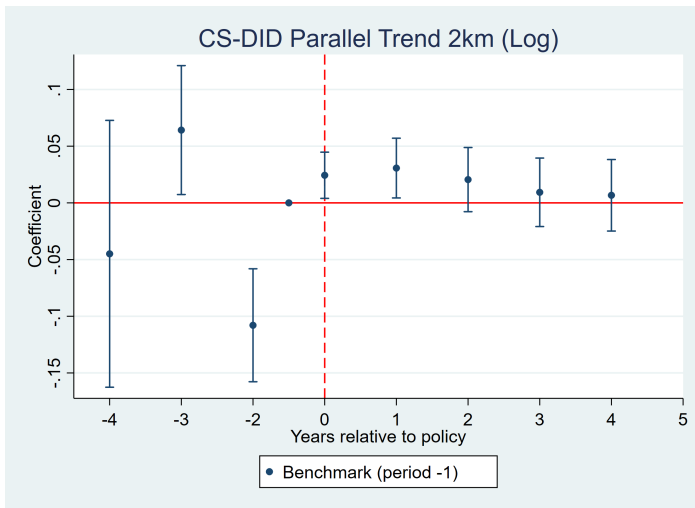
CS-DID: Not Yet Treated Control

Counts Level				Log Level			
	(1) 1km	(2) 2km	(3) 3km		(1) 1km	(2) 2km	(3) 3km
ATT	1.556*** (0.121)	4.715*** (0.328)	8.774*** (0.594)	ATT	0.129*** (0.011)	0.169*** (0.013)	0.187*** (0.014)
N	39,176	39,176	39,176	N	39,176	39,176	39,176

Event Study: CS-DID (Base)



Event Study: CS-DID (Log)



PPML: Baseline vs NTL Control

	<i>Baseline</i>			<i>+ Nighttime Lights</i>		
	(1) 1km	(2) 2km	(3) 3km	(4) 1km	(5) 2km	(6) 3km
DID	-0.021 (0.017)	-0.013 (0.015)	-0.006 (0.015)	-0.019 (0.016)	-0.010 (0.015)	-0.001 (0.015)
Hospital FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
NTL Control	No	No	No	Yes	Yes	Yes
N	147,853	156,693	159,764	147,853	156,693	159,764

PPML + Province Trend: Baseline vs NTL Control

	<i>Province Trend</i>			<i>Province Trend + NTL</i>		
	(1)	(2)	(3)	(4)	(5)	(6)
	1km	2km	3km	1km	2km	3km
DID	-0.012 (0.015)	-0.007 (0.013)	-0.003 (0.012)	-0.010 (0.014)	-0.004 (0.012)	0.001 (0.012)
Hospital FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Province Trend	Yes	Yes	Yes	Yes	Yes	Yes
NTL Control	No	No	No	Yes	Yes	Yes
N	147,853	156,693	159,764	147,853	156,693	159,764

Event Study: PPML (2km)

